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Double Deep Q-Network-Based Energy-Efficient Resource Allocation in Cloud Radio Access Network

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ABSTRACT Cloud radio access network (CRAN) has been shown as an effective means to boost network performance. Such gain stems from the intelligent management of remote radio heads (RRHs) in terms of on/off operation mode and power consumption. Most conventional resource allocation (RA) methods, however, optimize the network utility without considering the switching overhead of RRHs in adjacent time intervals. When the network environment becomes time-correlated, mathematical optimization is not directly applicable. In this paper, we aim to optimize the energy efficiency (EE) subject to the constraints on per-RRH transmission power and user data rates. To this end, we formulate the EE problem as a Markov decision process (MDP) and subsequently adopt deep reinforcement learning (DRL) technique to reap the cumulative EE rewards. Our starting point is the deep Q network (DQN), which is a combination of deep learning and Q-learning. In each time slot, DQN configures the status of RRHs yielding the largest Q-value (known as state-action value) prior to solving a power minimization problem for active RRHs. To overcome the Q-value overestimation issue of DQN, we propose a Double DQN (DDQN) framework that obtains optimal reward better than DQN by separating the selected action from the target Q-value generator. Simulation results validate that the DDQN-based RA method is more energy-efficient than the DQN-based RA algorithm and a baseline solution.

INDEX TERMS Cloud RAN, double deep Q-network (DDQN), energy efficiency (EE), Markov decision process (MDP), power allocation.

I. INTRODUCTION

The drastic increment of mobile subscribers and the demand for the ultra-high-speed mobile broadband network have accelerated the telecommunication sector towards the adoption of the fifth-generation (5G) communication system. Statistically, mobile users and data traffic are expected to reach nine billion and 71 exabytes in 2020, respectively [1]. Such a trend leads to the densification of base stations (BSs), which comes at the expense of high operating cost, CO₂ emission, and energy consumption. In fact, around 57% of the total energy consumption in cellular networks comes from BS [2]. Thus, improving the energy efficiency (EE) at radio access network (RAN) level has become a hot research topic [3], [4].

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In a typical RAN, a BS is physically connected to a fixed number of antennas, which limits the potential performance gain due to spatial correlation [5]. Cloud RAN (CRAN) [6] places antennas (known as remote radio heads (RRHs)) at different geographical positions within the cell. All RRHs are then connected to a centralized control unit known as a baseband unit (BBU) through a high-speed front-haul link. Such paradigm helps in meeting the traffic volume requirements in terms of stability, expandability, elasticity, and efficiency [7].

Inspired by the above advantages of CRAN, resource allocation (RA) for CRAN has been extensively studied. Total throughput [8], transmission power [9], and EE [10] are the prevalent RA objective functions which quantify the CRAN system efficiency. These RA works, however, do not consider the energy switching cost of RRHs during optimization. In fact, the overhead associated with the change of active/sleep mode can be significant [11]. Obviously, the

decisions of which cannot be solved directly in a slot-by-slot manner.

Markov decision process (MDP) can model the sequential decision-making RA problems, which can be effectively solved by reinforcement learning (RL) [12]. The principle behind RL is that an agent continuously learns and generates (sub)-optimal control actions based on interactions with environment. In the context of dynamic network environment, model-free RL algorithms such as Q-learning are often used to estimate the value function (known as Q-value function) of each state-action pair [13].

In this paper, we aim to optimize the EE subject to the constraints on per-RRH transmission power and user data rates. To this end, we formulate the EE problem as an MDP and subsequently adopt the deep RL (DRL) technique to reap the cumulative EE rewards. Our starting point is the deep Q network (DQN) [14], which is a combination of deep learning and Q-learning. In each time slot, DQN configures the status of RRHs yielding the largest Q-value (known as state-action value) prior to solving a power minimization problem for active RRHs [31]. To overcome the Q-value overestimation issue of DQN, we propose a Double DQN (DDQN) framework [15] that obtains optimal reward better than DQN by separating the selected action from the target Q-value generator. Simulation results validate that the DDQN-based RA method is more energy-efficient than the DQN-based RA algorithm and a baseline solution.

A preliminary version of this paper was presented at IEEE CCET 2020 [16]. This version is extension of that paper which focuses on EE maximization instead of power minimization. The main contributions of this work are summarized as follows:

1. We introduce DDQN to solve the EE optimization problem. Specifically, DDQN learns a near-optimal policy to configure the on/off modes of RRHs to improve the EE while meeting the user quality of service (QoS) requirements according to the environment. This is established by defining EE-oriented reward function, action space, and state-space.
2. In contrast to the DQN based scheme adopted in [31], we solve the EE optimization problem by using DDQN, which guarantees the stability.
3. Previous works [16], [31], and [35] rely on the static state-space feature to maximize the certain objective function. In this work, we optimize the objective function by considering the dynamic state-space feature, which is dynamically updated by the movement of UEs at each step.

Extensive simulation results are presented to validate the effectiveness of the proposed DDQN-based RA algorithm. It shows that the proposed solution is promising in handling time-variant dynamic problem.

The rest of this paper unfolds as follows. Related work for this research is presented in Section II. We describe the system model, followed by the problem formulation in

Section III. The proposed DDQN-based RA scheme is introduced in Section IV. The simulation parameters and results are discussed in Section V, followed by the conclusion and future work in Section VI and Section VII.

II. RELATED WORKS

Pursuing high EE is becoming the mainstream for future wireless communication design. This is especially relevant for CRAN, which is a promising architecture for the 5G network. Therefore, several researches focusing on optimizing the on/off RRH scheduling and/or transmission power have been proposed.

In [17], high-power BSs are replaced by low-power RRHs in a dense network distribution and it shows a great reduction in energy consumption. In [18], a mixed-integer non-linear programming problem is formulated for achieving good EE in energy harvesting aided heterogeneous-CRAN. In [19], the joint selection of RRH is considered for power minimization as a RA problem in the formation of group spares scattered for a green CRAN. In [20], a green CRAN is achieved by solving a joint RRH selection and power minimization beamforming problem. In [21], the effect of minimizing total energy consumption is studied in CRAN, showing that higher EE depends on the user's target demand. In [22], per-antenna transmission power is formulated to maximize the EE and proportional data rate among the users. The above work [17]–[22] formulated the RA problem for EE, joint cell activation, and user association as a convex optimization problem in the traditional model-based approach. However, these works do not consider the time-correlated scenario, where the actions in time slot t may influence future network utility. For example, it is shown that the energy switching cost of RRHs can be significant [11].

MDP can model the sequential decision-making RA problems, which can be effectively solved by a model-free RL framework. Unlike the model-based approach, the model-free approach maximizes the EE while satisfying the QoS demand of user equipment for the whole operational period in real-time. The advantages of RL are two-fold. On the one hand, the RL agent can generate (sub)-optimal control actions based on the current reward feedback from environment interaction. On the other hand, the RL agent can optimize the long-term network utility, which is essential in dynamic systems such as CRAN. Q-learning, one of the most popular RL algorithms, has been leveraged in solving the CRAN EE problem. In [23], a Q-learning based on/off policy is proposed to reduce power consumption while satisfying the user's data traffic demand. However, the constructed Q-table cannot scale well to a sizable state-action space.

A step forward from looking up the Q-value table in Q-learning is the Q-value approximation via deep learning. In [24], a DQN based dynamic RA for self-powered ultra-dense networks is proposed to enhance EE. Another DQN work in [26] is presented to study the tradeoff between spectral efficiency (SE) and EE. Similar framework can be found in [27], [28], where the environment settings are either

TABLE 1. Related work summary.

Ref	Approaches	Problem formulation	Solution	Limitation
[2][5][6] [9][11] [18][19]	Classic Approach	Model-based approach for energy-efficient resource allocation management.	Solved the resource allocation problem for energy efficiency based RRH selection, beamforming, and base station for minimization of power and energy following the traditional approach.	1.Relies on explicit objective functional expressions. 2.Cannot be guaranteed in complicated, high-dimensional scenarios.
[20][21] [25][36] [37]	Heuristic Approach	Model-based approach for energy-efficient,QoS improvement and delay minimization.	Solved the resource allocation problem for energy efficiency, improve QoS requirement of the user equipment and minimize the delay for the internet of things.	Less robust and cannot be proven rigorously using mathematics.
[7][8] [10] [22]	Programming Approach	Model-based approach for cell-activation and user association.	Solved the cell activation and user association joint optimization problem by using model-based approach in CRAN to balance the user QoS requirement and EE.	1.This method needs to be recalculated from the beginning of each iteration. 2.Some cases using programming algorithms rely on accurate predictions.
[16][24] [26][27] [28][31] [35]	Deep reinforcement learning (DRL)	Model-free approach based on reinforcement learning.	Solved the power allocation, energy efficiency and resource allocation in CRAN based on model free reinforcement learning approach.	1.Too much reinforcement learning can lead to an overload of states, which can diminish the results. 2.Reinforcement learning needs a lot of data and a lot of computation. It is data hungry.

vehicle-to-vehicle (V2V) or mobile edge computing (MEC). Some of the related work is summarized in Table. 1. However, the above [24]–[28] methods update the target Q-value using the same network parameters and can lead to the problem of overestimating the Q-value. This motivates us to leverage the DDQN framework which includes two Q functions to select and evaluate the action. It not only provides fast convergence to the Q-value function but also improves the EE of the proposed system.

III. SYSTEM MODEL

Fig. 1 depicts a downlink CRAN, where all digital signal processing (DSP) is done at BBU, which connects all RRHs. Each RRH compresses and forwards the received radio signal from BBU to user equipments (UEs) and vice versa. The BBU acts as an agent that continuously interacts with the environment and selects the action based on the input states from RRHs and UEs.

We consider a time period $\mathcal{T}$, which is divided uniformly into time slot t , represented as $\mathcal{T} = \{1, 2, \dots, T\}$. Each UEs position changes randomly and reports its channel state information (CSI) and data rate demands $D_u \in [D_{max}, D_{min}]$ to the BBU. The data rate demands D_u of each UE is available to the BBU pool. In a case where this information is not available, the service provider will report such information to the BBU cloud. The RL agent will then perform the RRH activation by sending the switching decision (on/off) to RRH

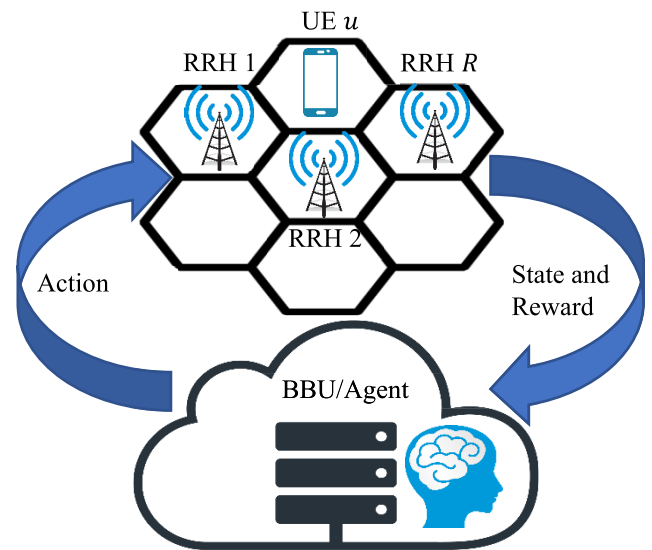


FIGURE 1. DRL-based C-RAN scenario.

and also monitors the UE mobility and its decision effect. Finally, the reward is calculated by the RL agent by combining the user satisfaction and power saving of all RRH in a centralized manner. For simplicity, we assume that all RRHs and UEs are equipped with a single antenna, which can be easily generalized in the case of multiple antennas [25].

A. NETWORK MODEL

We consider a set of RRHs $\mathcal{R} = \{1, 2, \dots, R\}$, a set of UEs $\mathcal{U} = \{1, 2, \dots, U\}$ and a set of BBUs $\mathcal{B} = \{1, 2, \dots, B\}$. The path loss of the system model is given by [31]:

$$PL(d_{r,u}) = 148.1 + 37.6 \log_2 d_{r,u} \text{ dB} \quad (1)$$

where $d_{r,u}$ represents the distance between the RRH r and UE u . The radio channel between the RRH and UE is modelled by [25]:

$$h_{r,u} = 10^{-PL(d_{r,u})/20} \sqrt{\varphi_{r,u} \varsigma_{r,u} n_{r,u}} \quad (2)$$

Specifically $\varphi_{r,u}$, $\varsigma_{r,u}$ and $n_{r,u}$ represent the antenna gain, shadowing and small-scale fading between RRH and UE, respectively. According to [25], RRHs cooperatively serve UE u through joint beamforming. In this way, the signal-to-interference-plus-noise ratio (SINR) of the UE u at time slot t , $\gamma_u(t)$ can be written as [25]:

$$\gamma_u(t) = \frac{|h_u^H(t) w_u(t)|^2}{\sum_{v \neq u} |h_v^H(t) w_v(t)|^2 + \sigma^2} \quad (3)$$

where $h_u(t) = [h_{1,u}, h_{2,u}, \dots, h_{R,u}]^T$ and each component $h_{r,u}$ represents the channel gain between RRH r and UE u at time slot t ; similarly $w_u(t) = [w_{1,u}, w_{2,u}, \dots, w_{R,u}]^T$ is the weighted vector and each component of $w_{r,u}$ represents the beamforming weight between RRH r and UE u at time slot t . σ^2 represents the background noise. The Shannon capacity for UE u at time slot t , $C_u(t)$ can be calculated as [25]:

$$C_u(t) = B \log_2 \left(1 + \frac{\gamma_u(t)}{\Gamma_m} \right) \quad (4)$$

where B is the transmission bandwidth and Γ_m denotes the realistic capacity gap depending on the modulation scheme.

B. POWER CONSUMPTION MODEL

Regarding the power consumption model, we consider four components [29]: (1) $p_{r,active}$: consumed power of active RRH; (2) $p_{r,trans}$: transmission power due to power amplifier; (3) $p_{r,switch}$: switching power of RRH; and (4) $p_{r,sleep}$: consumed power of sleep RRH. Let $\mathcal{M}$, $\mathcal{N}$ and $\mathcal{S}$ denote the set of active, sleep and mode-transition RRHs, respectively. Then, the power consumption can be expressed as [25]:

$$p_r = \begin{cases} \frac{1}{\eta} p_{r,trans} + p_{r,active}; & r \in \mathcal{M} \\ p_{r,sleep}; & r \in \mathcal{N} \\ p_{r,switch}; & r \in \mathcal{S} \end{cases} \quad (5)$$

where $p_{r,trans} = \sum_{r \in \mathcal{M}} \sum_{u \in \mathcal{U}} |w_{r,u}|^2$ indicates the transmission power of RRH r and η is a constant indicating the drain efficiency of the power amplifier. The total power p_{total} of RRH is updated at time slot t can be expressed as [25]:

$$p_{total}(t) = \sum_{r \in \mathcal{M}_t} \sum_{u \in \mathcal{U}_t} \frac{1}{\eta} |w_{r,u}(t)|^2 + \sum_{r \in \mathcal{M}_t} p_{r,active} + \sum_{r \in \mathcal{N}_t} p_{r,sleep} + \sum_{r \in \mathcal{S}_t} p_{r,switch} \quad (6)$$

C. SPECTRAL EFFICIENCY AND ENERGY EFFICIENCY IN CRAN

The spectral efficiency (SE) and EE are the key components for designing any wireless communication system. The SE represents the transmit information bits per hertz and can be computed as a sum of throughput C_u to the bandwidth of the system. Mathematically, SE can be expressed as:

$$SE(t) = \frac{\sum_{u=1}^U C_u(t)}{B} \quad (7)$$

Similarly, the EE represents the transmit information bits per joule and can be computed as a sum of throughput C_u to the total power consumption. Mathematically, EE can be computed as:

$$EE(t) = \frac{\sum_{u=1}^U C_u(t)}{B \times p_{total}(t)} \quad (8)$$

In particular, the EE will not be very high when the transmission power is uncontrolled even though it brings much high data rate. Therefore, it is necessary to develop an adequate power allocation transmission scheme to obtain higher EE.

D. PROBLEM FORMULATION

Green communication is the mainstream for future wireless communication design. Therefore, we consider EE as the network utility. Our objective is to maximize the long-term EE subject to constraints on per-RRH transmission power and user data rate. This is achieved by selecting the active set of RRHs in the time slot t and choosing the transmit power levels of the active RRHs. Excessive on/off switching of RRHs should be avoided due to the penalty of $p_{r,switch}$. Let us denote the overall beamforming weights by a matrix w with $[w]_{r,u} = w_{r,u}$. Then, the EE problem can be formulated as:

$$\max \sum_{t=1}^T EE(t) \quad (9a)$$

$$\text{s.t. } C_u(t) \geq D_u(t), \quad \forall u \in \mathcal{U}, \forall t \in \mathcal{T} \quad (9b)$$

$$\sum_{u \in \mathcal{U}_t} |w_{r,u}(t)|^2 \leq P_r, \quad \forall r \in \mathcal{M}_t, \forall t \in \mathcal{T} \quad (9c)$$

Constraints (9b) means that the user's data rate must be greater than or equal to the target data rate for each UE. Constraints (9c) specifies the upper limit of the RRH transmit power. Obviously, problem (9) is difficult to solve directly since the on/off decisions in adjacent time intervals are inter-related. Furthermore, traffic demands on cellular networks fluctuate in both the temporal and spatial domains. Fortunately, RL can effectively solve such problem and it motivates the design of the solution in next section.

IV. ALGORITHM DESCRIPTION

In this section, we present a DRL-based energy efficient RA framework to solve the problem of (9). We divide the problem into two subproblems: (1) RRH selection and (2) transmit

power minimization. All these optimizations are performed slot-by-slot by the BBU.

Almost all the problems of RL can be modeled as a Markov decision process (MDP) [39]. The MDP describes the behavior of environment for the RL agent. In our case, we consider a fully observable environment (FOMDP) for the RL agent. The main goal of MDP is to find an optimal policy to maximize the utility function. Policy is the mapping function from state space to action space. Therefore, before explaining the DRL approach, first we define the elements of MDP.

A. MARKOV DECISION PROCESS ELEMENTS

The MDP is a tuple of (S, A, G, S') , where S denotes the set of possible state, A denotes the action. At time slot t , the agent observes the state $s_t \in S$ and select an action $a_t \in A$. Based on the selected action, feedback is generated from the environment in terms of $G(s_t, a_t)$, and moves to the next state with a transition probability $P_r(S'|s_t, a_t)$. In this paper, we proposed a DDQN approach to obtain the optimal policy for RRH to turn on/off and maximize the EE in CRAN. However, to utilize the DRL approach for RL agent, we define the state-space, action-space, and reward function first as:

1) STATE-SPACE

At time slot t , the RL agent should know about all the user data rate demand $D_u(t)$, the status of all RRHs $v_r(t)$ and the channel between UE u and RRH r . So, the state space s_t is:

$$s(t) = \left[D_1(t), D_2(t), \dots, D_u(t), v_1(t), v_2(t), \dots, v_R(t), g_{1,1}(t), g_{1,2}(t), \dots, g_{R,U}(t) \right]^T \quad (10)$$

In equation (10), $v_r(t)$ has binary value and that indicates the on/off status of RRH, where $v_r(t) = 1$ implies that RRH r is turned on; otherwise, $v_r(t) = 0$, and $g_{R,U}(t)$ indicates the transmission rate between the RRH R and UEs U . Which is dynamically updated with the random movement of UEs.

2) ACTION-SPACE

At time slot t , we have restricted the RL agent to make a decision based on single RRH. After exploration and exploitation, the RL agent will be able to predict, which RRH to turn on or to turn off. Finally, the RL agent derives its action based on the active set of RRH $\mathcal{M}$. Note in each decision epoch; the action can affect the next state s' in the next time slot t .

3) REWARD

The reward shall reflect the EE improvement, which can be defined as:

$$G(t) = EE(t) = \frac{\sum_{u=1}^U C_u(t)}{B \times p_{total}(t)} \quad (11)$$

Iterative methods such as dynamic programming (DP) and Q-learning can be used to solve the above mentioned MDP

problem. However, DP requires a full information about the environment and reward function for solving MDP. This is especially challenging in the context of highly dynamic network conditions. Therefore, we resort to Q-learning, which can maximize the expected value of the accumulated total reward, without directly modelling the CRAN environment.

B. DEEP Q-LEARNING APPROACH

In Q-learning, the agent estimates the Q-value function $Q(s, a)$, where $a = \pi(s)$ and π is a specific policy. The system that starts in state s and follows the action a , it will reach the successor state s' and will continue to follow action $a' = \pi(s')$. Mathematically, $Q(s_t, a_t)$ can be expressed as [38]:

$$Q^\pi(s_t, a_t) = \mathbb{E}_\pi \left[\sum_{k=t}^T \mu^{k-t} G(k) | s_t, a_t \right] \quad (12)$$

where $G(t)$ represents the reward achieved in time slot t , and $\mu \in [0, 1]$ represents the discount factor. If $\mu = 0$, the agent only considers the current rewards, whereas if $\mu \leq 1$ the agent considers later rewards.

$Q^\pi(s_t, a_t)$ is essentially the long-term mathematical expected returns generated by MDP. This means that for all the actions selected by an optimal policy $\pi^*(s) = \text{argmax} Q(s_{t+1}, a_{t+1})$, such that $a_{t+1} = \pi^*(s_t)$ that maximizes the action-value function of (12) and satisfy $Q^{\pi^*}(s_t, a_t) > Q^\pi(s_t, a_t)$ for all Q-value function. According to Bellman's equation, Q-learning $Q(s_t, a_t)$ can be expressed as [14]:

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha [G(t) + \mu \max_{a'_{t+1}} Q(s_{t+1}, a'_{t+1}) - Q(s_t, a_t)] \quad (13)$$

where α is the learning rate. Conventionally, a Q-table is constructed to store all possible $Q(s, a)$ according to (13). However, the challenge lies in the convergence of Q-table when the state-action space grows. This problem is known as the curse of dimensionality [30]. To overcome the problem of dimensionality a better estimation alternative is to adopt a neural network which is known as DQN [14].

Let θ denotes the vector of the weights of a neural network. We define two Q-networks namely online Q-network $Q(s, a; \theta(t))$ and target Q-network $Q(s, a; \theta'(t))$ as shown in Fig. 2a. More precisely, $Q(s, a; \theta(t))$ is trained by adjusting the weights $\theta(t)$ at time slot t with respected to the target value $y(t)$ as [14]:

$$y(t) = G(t) + \mu \max_{a'} Q(s', a'; \theta'(t)) \quad (14)$$

where θ'_t is copied from the parameter θ_t at some previous iteration. Such technique known as network cloning removes correlations in the observation sequences [14]. Note that both networks stem from the same neural network, albeit a time delay. Combined with experience replay that relies on a random sample of past experiences, the learning stability can be further improved. Specifically, the DRL agent stores its experiences at each time slot t , $e_t = (s_t, a_t, G(t), s_{t+1})$ in a

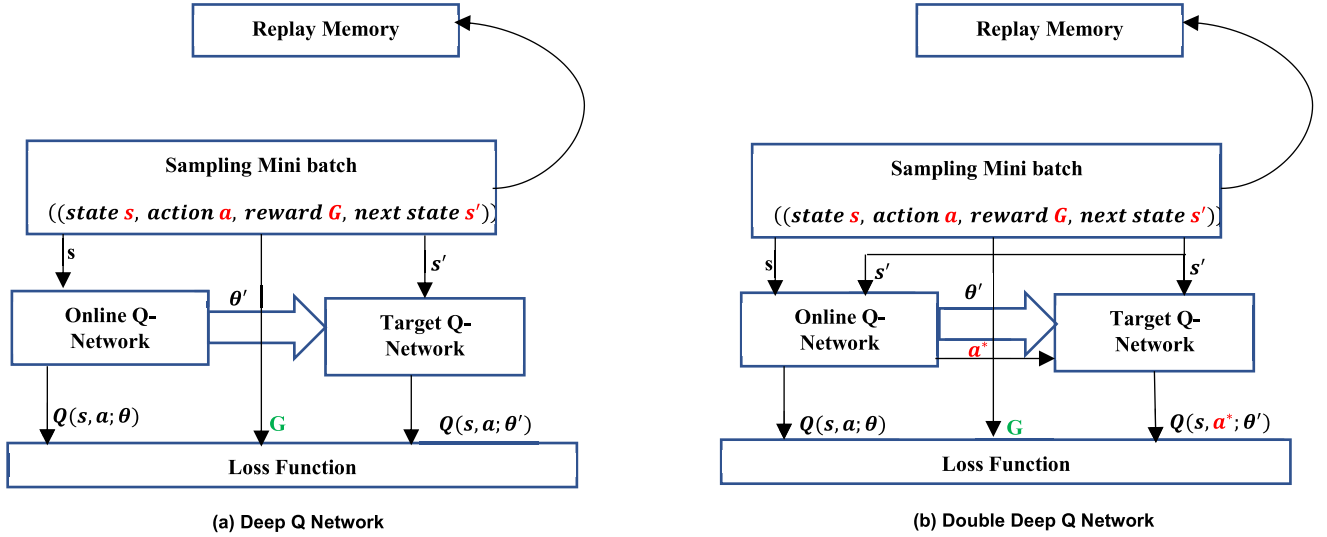


FIGURE 2. Flow of Q-network.

dataset $D_t = \{e_1, e_2, \dots, e_t\}$. A group of samples is collected from the set of stored samples, where the gradient descent of the mini-batch can be facilitated to minimize the loss function relative to the target value $y(t)$ in (14).

C. DOUBLE DEEP Q-NETWORK APPROACH

DQN have achieved beyond the human-level performance in many games. However, DQN uses single *max* mathematical estimator for the selection and evaluation of an action. Therefore, learning agents are sometimes confused about selecting or evaluating action, which leads to overoptimistic action values. These overestimations occurred due to positive bias from the *max* mathematical estimator used in DQN to update its Q-value function. The double Q-learning method was proposed to solve the overestimation action value problem [15], as shown in Fig. 2b. In DDQN, the action selection and action evaluation are decoupled into two separate *max* function estimators. This double estimator method gives fairer value, which results in no overestimation action values. For DDQN the target value of (14) can be written as [15]:

$$y(t) = G(t) + \mu Q \left(s', \underset{a^*}{\operatorname{argmax}} Q(s', a^*; \theta(t)); \theta'(t) \right) \quad (15)$$

At time step t , one set of weighted parameters θ is used to update the ϵ -greedy policy while other set of weighted parameters θ' determine the policy value. The ϵ -greedy policy represents the probability by which an agent takes a random action promoting exploration instead of an action determined by the maximum of the Q-value of the next state. To update Q-value function for i^{th} iteration of DDQN will remain the same as (12); however, the target value is changing as [15]:

$$y(t) = G(t) + (s', \underset{a^*}{\operatorname{argmax}} Q(s', a^*; \theta_i); \theta_{i-1}) \quad (16)$$

Note that the target network in both DDQN and DQN uses the previous iteration $i-1$ parameters. However, for generalization, any previous iteration $(i-k)^{\text{th}}$ parameters

can be used by the target network. Then the parameters of the target network are updated periodically with online copies of the network parameters.

D. TRANSMIT POWER ALLOCATION

Recall that DQN and DDQN accounts for RRH selection in each time slot t . Given the set of active RRHs $\mathcal{M}_t$ problem (6) can be reduced to slot-by-slot optimization.

$$\min_{\mathbf{w}_t} \sum_{r \in \mathcal{M}_t} \sum_{u \in \mathcal{U}_t} |w_{r,u}(t)|^2 \quad (18a)$$

$$\text{s.t. } \gamma_u(t) \geq \text{SINR}_u(t), \quad \forall u \in \mathcal{U}_t \quad (18b)$$

$$\sum_{u \in \mathcal{U}_t} |w_{r,u}(t)|^2 \leq p_{r,\text{transmit}}, \quad \forall r \in \mathcal{M}_t \quad (18c)$$

where $\text{SINR}_u(t) = \Gamma_m (2^{D_u(t)/B} - 1)$. Constraint (18b) implies that the demand of user u is guaranteed. Problem (18) belongs to a convex optimization problem as it can be transformed into a second order cone optimization problem [32]. Such problem can be solved efficiently via a conventional method [33]. Note that infeasible solution may arise due to insufficient active RRHs, especially in the early learning stage. In this case, we set $G(t) = 0$ which will render the agent become more aggressive to turn on RRHs in future time slots t . The pseudo-code for DDQN is explained in Algorithm 1. In this study, we implement a fully connected feed-forward neural network (FFNN) consisting of two-hidden layers with 64 and 32 neurons [31]. The input and output layers are connected to the hidden layer with a set of activation functions as a RELU. Furthermore, we establish an experience memory capacity $N_D = 500$ with a capacity of mini batch = 64 and $\mu = 0.995$.

V. PERFORMANCE EVALUATION

In this section, we analyze DDQN based RA performance in CRAN and compares it with simple DQN and traditional

Algorithm 1 DDQN Based Resource Allocation

1: Set $t = 1$
Initialize Experience memory N_D and soft update with τ .
Initialize main network with a random weight and biases as θ
Initialize the target network as a copy of primary network weights and biases as θ'
2: **for** each episode **do**:
3: Initialize state s_t in eq (10)
4: **for** each time slot **do**:
5: Select an action a_t based on ϵ -greedy policy
6: Obtain immediate reward $G(t)$ and observe next state s'
7: Solve (17) and obtain optimal beamforming solution
8: Store experience $(s_t, a_t, G(t), s'_t)$ into N_D
9: Randomly sample some minibatches $(s_t, a_t, G(t), s'_t)$ from N_D
10: Calculate the target Q-value in target deep network
11: **If** DRL = DDQN, set target

$$y(t) = G(t) + \mu \max_{a'} Q(s', a'; \theta'_t)$$

12: **If** DRL = DDQN, set target

$$y(t) = G(t) + \mu Q\left(s', \underset{a'}{\operatorname{argmax}} Q(s', a'; \theta(t)); \theta'(t)\right)$$

13: Train main network to minimize loss function $L(\theta)$:

$$L(\theta) = \mathbb{E} \left[(y(t) - Q(s_t, a_t; \theta))^2 \right] \quad (17)$$

14: Perform gradient descent step on

$$(y(t) - Q(s_t, a_t; \theta(t)))$$

15: Update target deep networks after some step as

$$\theta'(t) = \tau \theta(t) + (1 - \tau) \theta'(t)$$

16: $t = t + 1$
17: **end for**:
18: **end for**

approaches (Full coordinate association method (FA), where all RRH are turn on and user can be associated with multiple RRHs) by using Tensorflow Spyder IDE 3.3.4 [34]. The figures are based on the 100 testing episodes after successfully run for 1000 training episodes. To ensure the achievement of power-saving versus user data rate demand, we considered two different scenarios: (i) $R = 5$, $U = 2$ and (ii) $R = 12$, $U = 4$. To satisfy the fair comparison, we followed the same system configuration as [35], which is tabulated in Table 2. For a long-term RA optimization, we considered dynamic channel gain with dynamic UE demand. The UE demand is uniformly distributed in the interval of [10, 70] Mbps. Note we considered each decision epoch to a minimum of 5 seconds. The UE data rate demand and channel gain changes after each decision epoch. We then calculated the total average power consumption for a 7 minutes timeslot interval. For Fig. 3, we assumed that the resource occupied from the given set of RRHs to UE are dynamically changing

TABLE 2. Simulation parameters.

Parameter	Symbol	Value
Noise power	σ^2	-102 dBm
Bandwidth	B	10 MHz
Active power	$p_{r,\text{active}}$	6.8 W
Transmit power	$p_{r,\text{transmit}}$	1 W
Sleep power	$p_{r,\text{sleep}}$	4.3 W
Transition power	$p_{r,\text{switch}}$	3 W
Antenna Gain	$\phi_{r,u}$	9 dBi
Shadowing coefficient	$\varsigma_{r,u}$	8 dB
capacity gap	Γ_m	1
Small scale fading	$\eta_{r,u}$	$\mathcal{CN}(0,1)$
Power amplifier efficiency	η	25 %
epsilon-greedy policy	ϵ	0.05

at each time slot t with static user demand. However, from the Fig. 3, one can observe that the power is monotonically increased with increasing user demand. When the user demand is 10Mbps, DQN saves 10.39% more power than the FA approach, while DDQN saves 13.43% more power than DQN.

In Fig. 3, it can also be observed that at all points DRL approach outperform the FA method. Similarly, DDQN exceeds the performance of DQN at all points and saves 7-20% more power. As the UE demand is approaching to higher data rates, all the three approaches cannot satisfy the power consumption beyond 60Mbps due to resource bottleneck. As we have seen resource bottleneck problem occurs when the user demand increases beyond 60Mbps in earlier in Fig. 3. This is because when accommodating higher user demands, more RRH should be turned on most of the time. Thus, we increase the number of RRH to 12 with multi UE, as shown in the Fig. 3, our proposed DDQN based method is the evidence of consuming less power at all the points. When user demand is 10Mbps, DQN based method saves 7% more power than the FA approach. Similarly, DDQN saves 8% more power than DQN at the same point. However, power is increasing linearly with increasing user demand. When the user demand approaches to 70Mbps, DDQN saves 9.38% more power than the DQN approach. In contrast, DQN saves 7.69% more power than the FA method.

Later, we considered a more realistic scenario where both the resource occupied from given RRHs to the user and UE data rate demand dynamically changes at each time slot t . As mentioned earlier, we set the decision epoch to a minimum of 5 seconds. Such that UE data rate demand and resource occupied changes at each decision epoch. In Fig. 4, it can be observed that the fluctuation occurs at each time slot t with the changing UE data rate demand. From Fig. 4, we can see that DDQN consistently outperforms the DQN and FA approach in terms of power-saving and satisfying user's QoS requirements. These performances justify the effectiveness of our proposed method in a highly

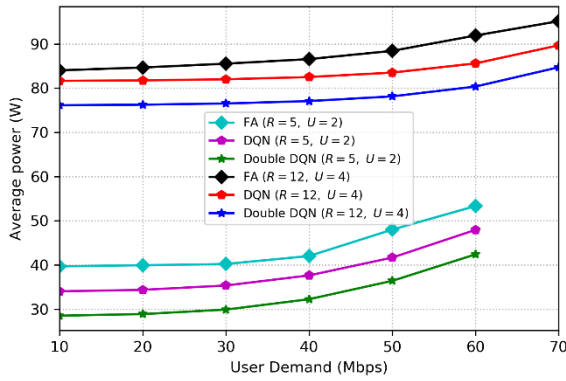


FIGURE 3. Power consumption vs. user demand.

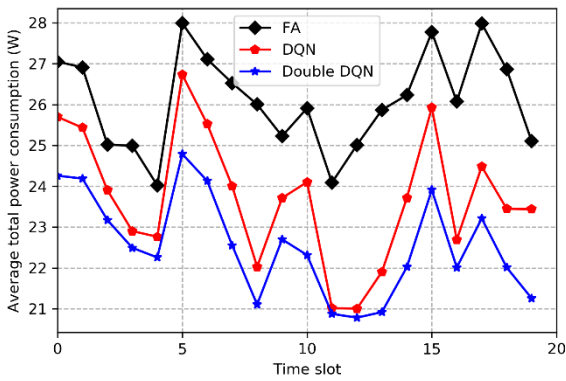


FIGURE 4. Power consumption vs. time slot.

dynamic case. In Fig. 5, energy efficiency is plotted against growing user demand. In this case, the increase in user demand may cause a linear increase in energy efficiency. The algorithm proposed by the DDQN exceeds the performance from DQN and FA scheme [25]. As seen in the Fig. 5, energy efficiency performance with DRL is better than the FA approach. The reason comes from the effect of considering past learning experiences. All the optimal RA for the FA approach relies on the instantaneous network state, which gives the slow convergence value. We can also observe from the Fig. 5, that our proposed method DDQN outperforms the energy efficiency around 7-18% at each point of increasing user data rate demand compare to DQN [31]. The reason comes from the effect of considering two separate Q-value estimators, that helps the agent to select an unbiased Q-value to estimate the action chosen using the opposite estimator. Noticeably, with increasing user demand, all three approaches for EE are increasing monotonically. However, when the user demand approach to 60Mbps, all three methods become constant, which no longer varies with growing user demand. This is because this scheme aims to maximize the utility function defines as the logarithmic function of the user's data rate, which requires more transmission power resulting in lesser energy efficiency. To mitigate the earlier problem, we increase the number of RRHs and UEs as shown in the Fig. 5. Due to the varied number

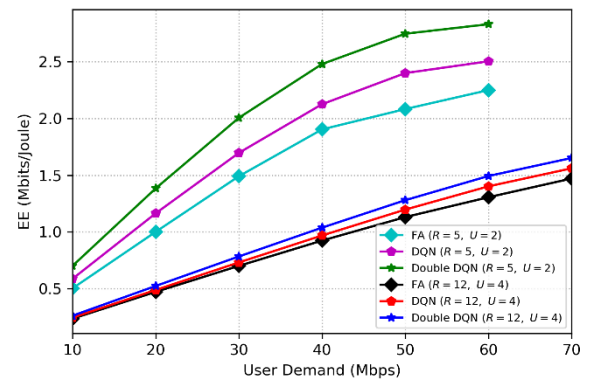


FIGURE 5. Energy efficiency vs. user demand.

of RRHs and users, the impact is observed on the energy efficiency. However, the performance of energy efficiency with the DDQN algorithm is outperforming compared to other approaches. The optimal energy efficiency increases with increasing user demand. This is because a larger number of RRH is active to obtain the optimal energy efficiency at each point of user demand. From Fig. 5, we concluded that increasing R and U affect energy efficiency in a complicated manner. Therefore, adjusting these parameters is essential to obtain higher energy efficiency. DDQN takes the advantage of two separate Q-network to turn on/off the RRH to the given users. Therefore, at each point, DDQN obtained 6-16% more energy efficiency than other approaches. In Fig. 6, we plotted energy efficiency versus total power consumption. According to the Fig. 6, energy efficiency increases linearly with the amount of maximizing power. However, as the power consumption reaches a certain level of the threshold $P_{\max}$ value, the energy efficiency achieves from all three approaches become constant with no increment of maximizing total power consumption. This is because total power $p_{\text{total}}(t)$ becomes greater than $P_{\max}$ value. From Fig. 6, we can also observe that DDQN is consistently outperforming two approaches. The maximum energy efficiency obtained for DQN approaches to 2.532Mbits/joule with an amount of power consumption of 47 W. While at the same time DDQN achieves the energy efficiency more than 2.831Mbits/Joule with a maximum power consumption of 42.3W. This increment satisfies the objective function defines in (8). In Fig. 7, energy efficiency is plotted against multiuser power consumption. Initially, energy efficiency for all three approaches increases slightly over a small increase in power. The energy efficiency obtained by all three approaches starts to decline after reaching to a maximum value. This is because all three approaches aim to maximize the user data rate's utility function, which requires higher transmit power, resulting in lower energy efficiency. From Fig. 7, DDQN achieves a maximum of energy efficiency 1.62Mbits/Joule with a power consumption of 84.92W. In contrast, DQN and FA approach achieve a maximum of energy efficiency 1.579Mbits/Joule and 1.512Mbits/Joule with an amount of power consumption of 89.21W and 94.982W respectively.

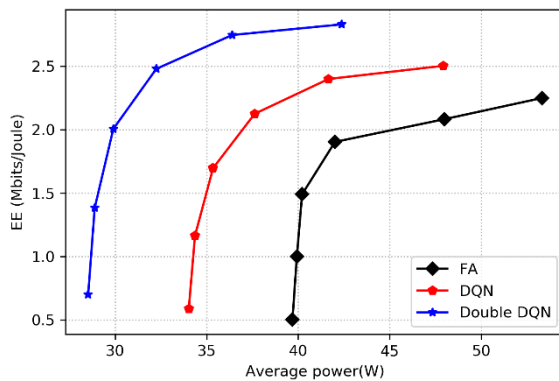


FIGURE 6. Energy efficiency vs. power consumption of $R = 5, U = 2$.

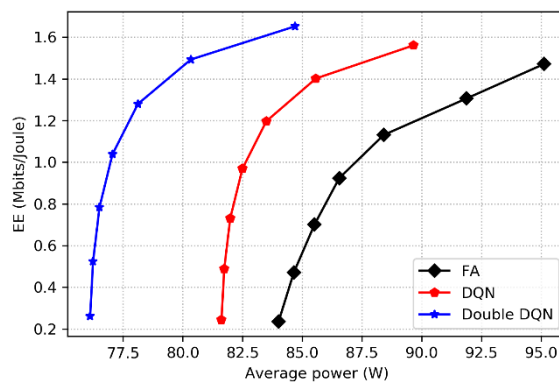


FIGURE 7. Energy efficiency vs. power consumption of $R = 12, U = 4$.

These performance shows the effectiveness of DDQN, that using the power allocation optimization approach, the system's performance is improved in terms of energy efficiency.

VI. CONCLUSION

In this paper, we have investigated a Double DQN based RA framework in CRAN that maximize the total energy efficiency subject to the constraint on per RRHs transmission power selection and user rates. Our starting point is the traditional approach where the reward function is achieved from the immediate action with ignoring its effect in the future. DQN is proposed using the past learning experience and considering the future effects based on the current action decision. However, the action overestimation problem gives the lower probability limit to estimate the maximum Q-value. Thanks to this framework, Double DQN separates selected action from the target Q-value generation leading to higher value of energy savings at CRAN. Extensive simulation results confirm the effectiveness that the Double DQN based RA method not only saves 22% more power performance over benchmarks but also improves the performance of EE of 20% with a minimum requirement of user data rates.

FUTURE WORK

In this work, we considered the function approximation method to approximate the dynamic state-space feature,

which easily converge to Q-value function. In the future, we will consider some more complex feature in the state-space, i.e., the latency and inaccurate channel state information. We will solve the complex state-space problem by using K-mean clustering and anchor graph hashing method.

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resource allocation optimization, machine learning, deep reinforcement learning, and 5G cellular networks.



Things (IoT), machine learning, deep learning, deep reinforcement learning, and video communications.



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